

| Question |     |       |  | Marking details                                                                                                                                                                                                           | Marks Available |     |     |       |       |      |
|----------|-----|-------|--|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|-----|-----|-------|-------|------|
|          |     |       |  |                                                                                                                                                                                                                           | AO1             | AO2 | AO3 | Total | Maths | Prac |
|          |     | (ii)  |  | 1.2 $\mu\text{m}$ = 3 marks<br>1.2 (no units) = 2 marks<br>(3.6-1.2) / 2 = 1 mark                                                                                                                                         |                 | 3   |     | 3     | 3     |      |
|          |     | (iii) |  | Any 2 x (1)<br><br>The temperature of solution (1)<br>pH of solution (1)<br>same type of muscle (1)<br>Same concentration solution (1)<br><br>OR 2 controlled variables without solution (1)                              |                 |     | 2   | 2     |       | 2    |
|          |     | (iv)  |  | Any 2 x (1) from:<br><br>• no overlap between actin and myosin filaments (1)<br>• no cross-bridges form / no possibility of myosin head moving actin (1)<br>• no contraction possible / sarcomeres cannot get shorter (1) |                 |     | 2   | 2     |       |      |
|          | (c) | (i)   |  | each muscle contracts to 'pull' in each direction / as one contracts the other relaxes to move the limb (1)                                                                                                               | 1               |     |     | 1     |       |      |
|          |     | (ii)  |  | Any 2 x (1) from:<br><br>Quadriceps contract (1)<br><br>The osteoarthritic group has a lower contraction force ORA (1)                                                                                                    | 1               | 1   |     | 2     |       |      |